Networking Data Issues and the Internet Checksum

proj 2: TCP - UDP, broadcast, etc.

Come to office hours (Ethernet, IPv4)

1. Data in Networks
   - just bytes
2. Size of Data
   - 8, 16, 32, 64
3. Bitmasking
4. Little vs. Big

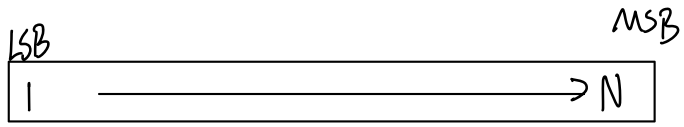
Treat as Array, not String

- use memcpy for moving data

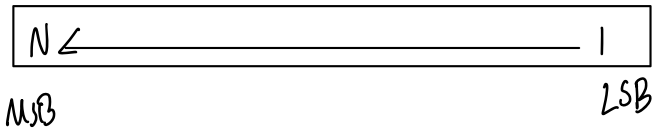
IP address = uint32_t

unsigned char = uint8_t

Little Endian



Big Endian



Almost all machines are little-endian because of intel

ntohs - network-to-host short

ntohl - network-to-host long

htons - host to network short

htonl - host to network long

→ applications handle endianness

Internet Checksum

in_cksum()

- unsigned short in_chksum (unsigned short * addr, int len)
  - typecast array to short

To check for errors in the IP header:

run in_cksum on the IP header

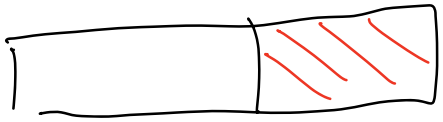
Two cases:

returns 0, no error

returns ≠0, error detected

Bit Masking:

IP[0] & 0x0F =



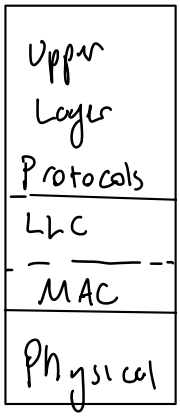
Take #, multiply by 4

LANs - defined by IEEE 802

Physical layer

Data Link layer

- Medium Access Control (MAC)
- logical link control (LLC)



The hope is standard changes don't destroy the network.

LLC
- frame assembly

MAC
- encoding / decoding

Ethernet Frame
- Address + type + Payload + FCS
- Physical layer carries Ethernet Frame (PDU)
  - preamble
  - start of frame delimiter (SFD)

Ethernet II and 802.3

Type < 1536 - 802.3

Type >= 1536 - Ethernet II
- ethertype = Ethernet II

Layer 2 PDU: frame

Max payload

normal: 1460

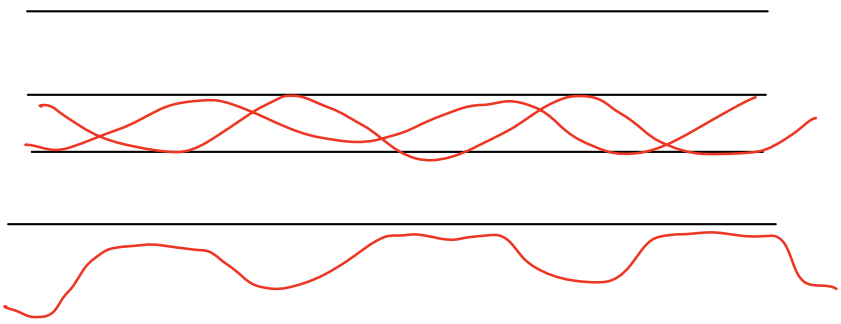
jumbo: 9000 (data centers)

Multimode fiber

Single Mode fiber

modal dispersion in multimode fiber
- creates intersymbol inference in fiber transmissions

Multimode – core is thicker
light can bounce around more



↑ skewing of bits

(intersymbol interference)

single-mode fiber (thinner core)

- data doesn't bounce around at all
- no modal dispersion

Speeds of 800 GB/s

↳ 10s of KM

Wavelength division multiplexing

802.11 = Wi-Fi (IEEE Standard)

↳ Trade-name

802.11b → 802.11ax (Wifi 6E)

802.11a sucked!

- only 5c

WAP (Wireless access point)

- Base station
- Wireless Router
- Wireless Adapter

SSID - service set identifier